ZCAS UNIVERSITY

CIT2241 BUSINESS INFORMATION SYSTEMS

FINAL EXAMINATIONS

TUESDAY, 2nd JUNE 2026

09:00-12:00 HOURS

TIME ALLOWED: THREE HOURS (plus 5 minutes to read through the paper)

INSTRUCTIONS TO CANDIDATES:

1. **SECTION A:** This section is compulsory. You must answer **Question 1.**
2. **SECTION B:** Attempt **any three (03)** questions out of the **four (04)** questions provided.
3. Time allowed: **3 Hours plus five Minutes to read through the paper.**
4. This examination paper carries a total of **100 marks.**
5. Candidates **MUST NOT** turn this page **until** the Invigilator tells them to do so.

SECTION A: This section is compulsory. You must answer Question ONE.

QUESTION ONE

CASE STUDY

A large provincial hospital in Kitwe has been operating for many years using a fully manual patient record system. Patient files are stored in physical cabinets across different departments such as outpatient, inpatient, laboratory, pharmacy, and maternity. Over time, the hospital has experienced rapid growth in patient numbers, but its record-keeping processes have not evolved at the same pace.

As a result, the hospital is now facing several operational challenges. Patient files are frequently misplaced or duplicated, leading to delays in diagnosis and treatment. In some cases, clinicians are forced to wait for long periods while records are being manually retrieved from archives. There are also instances where patients' historical medical information is incomplete or inconsistent, affecting the quality of care provided. Administrative staff spend significant time searching for files instead of focusing on service delivery.

The hospital management has also raised concerns about reporting and decision-making. Aggregating data for monthly or annual health reports is time-consuming and often prone to errors. This makes it difficult for management to accurately track disease trends, resource usage, or staffing needs. Furthermore, storage space for paper records is becoming increasingly inadequate, and the risk of damage from fire, floods, or deterioration is a growing concern.

In response to these challenges, the hospital management has decided to replace the manual system with a computerized patient record system. The proposed system is expected to improve efficiency, enhance data accuracy, and support faster clinical decision-making. It will also enable authorized staff to access patient information more quickly and securely across departments.

However, the management team is aware that such a transition is complex and carries significant risks if not properly managed. They are concerned about system failure, resistance from staff who are accustomed to manual processes, data migration errors, inadequate training, and potential cybersecurity threats. There is also the challenge of selecting appropriate technology that aligns with the hospital's size, budget constraints, and long-term strategic goals.

To address these concerns, the hospital intends to implement the project using a structured and carefully planned approach. This includes conducting detailed needs assessment, engaging stakeholders from all departments, evaluating suitable technology options, and adopting a formal

systems development methodology. The management also plans to ensure staff training, change management, and phased implementation to minimize disruption to hospital services.

Ultimately, the goal is not only to digitize patient records but to transform the hospital's overall information management capability, ensuring improved patient care, better coordination among departments, and more reliable data for strategic healthcare planning.

Required:

a) Using the Systems Development Life Cycle (**SDLC**) approach, briefly describe stages involved in developing the new hospital system. **(5 marks)**

b) The hospital manager wants Excel to automatically display:

- "Busy" if the number of patients attended to is above 20
- "Not Busy" if the number of patients attended to is 20 or below
- Total Patients attended in a week
- Average Number of Patients attended to per day
- Highest Number of Patients attended to in a day

Write appropriate Excel formulas for the following tasks: **(10 marks)**

c) Which **FIVE (05)** factors do you suggest the hospital must consider during technology acquisition. **(5 marks)**

d) Using the **RAKID** and **DIKW** model, explain how knowledge can be managed in the hospital environment. **(5 marks)**

e) Analyze the hospital's need for information system using Anthony's Triangle. **(5 marks)**

f) Describe **FIVE (05)** risks associated with computerized information systems in hospitals. **(5 marks)**

g) As a BIS expert, advise **FIVE (05)** measures the hospital can implement to improve information security. **(5 marks)**

Total (40 marks)

QUESTION TWO - 20 MARKS

S CASE STUDY

A retail company, ***Source IT Solutions,*** in Lusaka has been collecting large amounts of sales data every day. However, management complains that despite having a lot of data, they are failing to make good business decisions. Also, the company discovers a data breach affecting 10,000 EU

customers on Friday at 5 PM. The IT team says they need until Monday to fully understand the scope. GDPR requires notification within 72 hours. Your legal team says waiting could mean a €20M fine, but notifying prematurely might cause unnecessary panic and you're not sure if personal data was actually accessed. The CEO is unreachable on a weekend retreat in Siavonga. What do you do and why?"

Required:

a) Discuss ***Source IT Solutions*** Legal obligation versus complete information and Risk of premature versus delayed notification. **(5 marks)**

b) Using the **Data-to-Information concept**, explain how raw data can be transformed into useful information in the company. **(5 marks)**

c) How can ***Source IT Solutions*** appreciate the importance of **Information Quality** in its decision-making. **(5 marks)**

d) Using the **DIKAR model**, explore how ***Source IT Solutions*** move from data. **(5 marks)**

Total (20 marks)

QUESTION THREE - 20 MARKS

CASE STUDY

A small Kantemba grocery shop in Lusaka's Mtendere has been recording sales and stock manually in books. The owner now wants to start using computers to improve business operations.

Required:

a) Help the owner to critically understand Business Information System (BIS). Provide five points in your answer. **(5 marks)**

b) Advise the Kantemba owner any **five (05)** benefits of using BIS in the grocery shop. **(5 marks)**

c) Suggest **five (05)** types of information systems the shop owner can use. **(5 marks)**

d) Explain **five (05)** challenges the business may face when introducing BIS. **(5 marks)**

Total (20 marks)

QUESTION FOUR - 20 MARKS

CASE STUDY

Masambililo College recently introduced a new student management system. Some lecturers are worried about unauthorized access to student results, while management wants the system to support decision-making at all levels. There is a huge debate whether to encapsulate or to encrypt the college data and how the college will remain safe from cyber-attacks.

Required:

a) Using the **CIA Triad, Repudiation and Non-repudiation**, explain how Masambililo College can protect student information. **(5 marks)**

b) With examples, identify the main categorization of Security Threats and describe the structured and unstructured type of threats. **(5 marks)**

c) As a BIS expert, distinguish **Encapsulation** from **Encryption** in information systems or software development. **(5 marks)**

d) Help the College Principal understand these terms; "**Evil Twin**", "**Script Kiddies**", "**Hackitivists**" "**DDOS attack**", "**Phishing**". **(5 marks)**

Total (20 marks)

END OF EXAMINATIONS